DormiSpace Application

BY

MISS Chanisara Jonsomjid 6588033
MISS Phanphum Prathumsuwan 6588081
MISS Achiraya Mankham 6588183

ADVISOR

Dr. Pawitra Liamruk

A Senior Project Submitted in Partial Fulfillment of the Requirements for

THE DEGREE OF BACHELOR OF SCIENCE (INFORMATION AND COMMUNICATION TECHNOLOGY)

Faculty of Information and Communication Technology
Mahidol University
2025

COPYRIGHT OF MAHIDOL UNIVERSITY

ACKNOWLEDGEMENTS

The authors would like to express their sincere gratitude to Dr. Pawitra Liamruk, our project advisor, for her continuous guidance, valuable advice, and encouragement throughout the development of this project. Her insightful suggestions greatly contributed to improving the quality and direction of our work.

The authors also wishes to express appreciation to Mr. Brian Marcar, who kindly reviewed and provided feedback on the English language and presentation of this proposal. Their support helped enhance the clarity and readability of the report.

The authors would also like to thank all participants who responded to our user requirement survey. Their feedback provided essential insights that helped us design a system aligned with real user needs.

We are deeply grateful to our team members for their dedication, cooperation, and teamwork, which made the progress of this project possible.

Finally, we would like to extend our appreciation to Mahidol University, Faculty of Information and Communication Technology, for providing the facilities, resources, and supportive environment necessary to complete this project successfully.

Miss Chanisara Jonsomjid

Miss Phanphum Prathumsuwan

Miss Achiraya Mankham

DormiSpace Application

MISS Chanisara Jonsomjid 6588033 ITCS/B

MISS Phanphum Prathumsuwan 6588081 ITCS/B

MISS Achiraya Mankham 6588183 ITCS/B

B.Sc. (INFORMATION AND COMMUNICATION TECHNOLOGY)

PROJECT ADVISOR: Dr. Pawitra Liamruk

ABSTRACT

Managing dormitories manually often leads to inefficiencies such as billing errors, delayed payments, and miscommunication between landlords and tenants. To address these issues, this project proposes the development of DormiSpace, a smart dormitory management application designed to centralize and automate key management processes. The objective of DormiSpace is to simplify room registration, rent payment, utility tracking, and maintenance requests within a single integrated platform.

The development process follows the Agile Scrum methodology to allow iterative design and user feedback. A user requirement survey involving 27 participants (20 tenants and 7 landlords) was conducted to identify essential features. System analysis and design were carried out using UML diagrams, functional decomposition, ERD, and DFD models. The planned implementation will employ Flutter for the frontend and Spring Boot with a PostgreSQL database for the backend, integrating an OCR module for meter reading and the Omise API for secure payments.

Survey results reveal strong demand for transparency, automation, and digital convenience in dormitory management. The expected outcome of this project is a functional prototype that improves operational efficiency, enhances landlord–tenant communication, and provides a reliable, user-friendly digital solution for modern dormitory management.

PAST TENSE

KEYWORDS : Dormitory Management System, Smart Application, Optical Character Recognition (OCR), Digital Payment, Flutter, Spring Boot

CONTENTS

Page

ACKNOWLEDGEMENTS ii
ABSTRACT iii

ABSTRACT 2
INTRODUCTION 5
1.1 Motivation 5
1.2 Problem Statement 6
1.3 Objectives of the Project 7
1.5 Expected Benefits 7
BACKGROUND 9
RELATED WORKS 11
Comparison of Existing Applications 12
METHODOLOGY 24
EVALUATION AND RESULTS 30
CONCLUSION AND FUTURE WORK 31
REFERENCES 32
AI USAGE ACKNOWLEDGMENT STATEMENT 34
APPENDIX A 35

CHAPTER 1

INTRODUCTION

This chapter presents the motivations, problem statement, objectives, scope and expected benefit of developing DormiSpace: A Dormitory Management Application. It discusses the increasing demand for affordable rental housing and the inefficiencies of manual dormitory management. The system aims to centralized digitalize operations, help landlords and tenants dealing with the documents, payment, and room maintenance and enhance overall management efficiency.

1.1 Motivation

In recent years, renting dormitories has become an increasing popular choice among university students and office workers due to the increasing prices of condominium rentals. Dormitory businesses play an important role in providing affordable and convenient options, especially in areas located near universities and industrial zones. These businesses make income from various sources such as room rental fees, utility, and facility, which are usually managed on a monthly basis.

However, a lot of small-scale landlords in Thailand continue to rely on traditional management methods including spreadsheets, handwritten contracts, and paper-based recordkeeping. These manual processes frequently result in time-consuming landlords effort, ineffectiveness, and inaccurate data. Tenant payment summaries, bill preparation, and electricity and water usage must all be done by hand by the landlord. As a result, as the number of tenants increases, dorm management becomes increasingly difficult and inefficient.

The case studies that provided motivation for this project was Chanisara APT, a family-run rental business located in Rayong, Thailand, where one of our group members owns a dorm for employees and students. Without an automated system, the landlord currently manages twenty units apartment buildings. The manual handling of all tenant contracts, payment and maintenance requests makes it difficult to effectively track payments, keep an eye on occupancy,

and handle repair requests. These problems slow down communication between landlords and tenants in addition to adding to the landlord's duty.

The majority of tenants, in their opinion, prefer to have access to real-time data, including monthly bills, maintenance status, and payment histories. They also expect modern conveniences such as online payment options and digital receipts instead of physical paper documents.

With the goal to address all requirements, the DormiSpace Room Management System was created, which offers a centralized and automated platform for managing dorm management. The system will make it easier for landlords to handle contracts, keep track of availability of rooms, and keep an eye on maintenance requests. This system supports the digital transformation of small dormitory businesses in Thailand by reducing manual work, reducing errors, and improving the experience for both landlords and tenants.

1.2 Problem Statement

The current management of DormiSpace is entirely manual, creating several operational issues. Additionally, these issues slow down service delivery, increase the workload for the manager, and reduce overall tenant satisfaction:

- **Disorganized and Insecure Contract Management:** All tenant contracts are physical paper documents stored without a proper organizational system. This method is not only inefficient but also poses a high risk of documents being lost or damaged, which could lead to legal and financial complications.
- **Lack of Real-Time Information:** Room availability is not tracked in real-time. To determine if a room is vacant or occupied, management must perform a manual check, which is slow and can lead to missed rental opportunities or conflicting information.
- **Inefficient Maintenance Handling:** Tenants report maintenance issues through various channels, including phone calls and LINE messages. This makes it difficult for management to track all requests effectively, leading to potential delays or oversights in addressing necessary repairs.

- **No Centralized Data System:** The absence of a unified system means that all critical rental data is scattered across physical papers and potentially different digital files, making comprehensive oversight and management difficult.

1.3 Objectives of the Project

The primary goal of the DormiSpace project is to develop a centralized rental management system that effectively addresses the identified operational challenges. The success of the project will be evaluated based on its ability to achieve the following key objectives:

1. To establish an integrated platform that enables landlords to efficiently manage tenant information, monitor utility consumption, and maintain accurate billing records with minimized human error.
2. To implement automated tools for utility data collection and bill computation that enhance operational accuracy, reduce manual workload, and streamline the overall management process for landlords.
3. To deliver a user-centric mobile application for tenants that provides real-time visibility of utility usage and supports secure digital bill payments, including access to detailed payment histories and electronic receipts.

1.4 Scope of the Project

The scope of this room management system is clearly defined to ensure the project can be completed effectively, while recognizing that it forms part of a larger potential platform in the future. The main functional areas within the project scope are as follows:

1. Room and Tenant Database: The system will support creating, editing, and deleting room and tenant records.

2. Contract Management: The system will include the capability to upload, store, and manage digital tenant contracts securely.

3. Maintenance Request Tracking: The system will allow tenants to submit maintenance requests online and enable management to track the status of each request.

1.5 Expected Benefits

There are some obvious and expected benefits like the proposed DormiSpace Room Management System will bring several clear and measurable advantages to the business. In addition, beyond the measurable savings and efficiency improvements, the system also offers important qualitative benefits that can strengthen the business. Some more details are as below:

- Reduced Printing and Storage Costs
- Less Administrative Time Required for Record-Keeping and Request Tracking
- Faster Turnaround for Maintenance Requests
- Improved Tenant Satisfaction and Trust Through Faster Service
- Better Decision-Making Due to Accurate, Real-Time Data
- Professional Image That Can Attract New Tenants and Improve Competitiveness

CHAPTER 2
RELATED WORKS

This chapter explores existing rental and dormitory management applications and studies, with a particular emphasis on technology-driven solutions supported by digital tools and platforms. It focuses on the essential features, benefits, and limits of current systems while exposing the gaps that small-business landlords and tenants continue to encounter. In addition, this chapter provides an overview of the background and current management operations of "Chanisara Apartment," the case study dormitory. It discusses the increased demand for low-cost long-term apartment rentals, as well as the operational issues that the company has as it continues to use traditional management techniques.

2.1 Background of Our Case Study

One of our group members' family businesses manage rental properties known as Chanisara Apartment, located in Rayong Province, Thailand. The business provides long-term dormitories for university students, office workers. It has grown to be a popular choice for local tenants due to its quiet environment, convenient location nearby industrial and educational areas, and reasonable prices.

Chanisara apartment consists of 20 fully furnished rooms designed for comfort and convenience. Each room is equipped with essential furniture and appliances, including a bed, wardrobe, desk, dining table, air conditioner, water heater, refrigerator, and a sink. Pets are not allowed, ensuring a clean and safe living environment. The rental price is 4,500 Baht per month, with a security deposit of 5,000 Baht, making the total initial move-in cost 9,500 Baht. The minimum rental contract is 6 months, while utility charges are calculated at 8 Baht per unit for electricity and 18 Baht per unit for water, with a minimum water charge of 100 Baht.

The apartment benefits from a great location near education and industrial estates. It is close to the BYD factory, Lotus Nikom Phatthana, Marie Wittaya School, King Mongkut's University of Technology North Bangkok (Rayong Campus), and several industrial estates, including CPGC, CPPC, and Pinthong 6. This accessibility makes it an attractive option for tenants who work or study in these areas.

Despite offering quality accommodations and satisfactory services, Chanisara Apartment relies on traditional management methods. Task Tenant move-in, contract preparation, maintenance tracking, and payment are among the manual tasks that are carried out via spreadsheets, paper documents, and telephone conversations. Inefficiencies based on these manual processes may include trouble finding records, a delay in maintenance response, and inaccurate monthly bill calculations. These processes have become increasingly time-consuming and challenging to effectively manage as the number of tenants increases.

2.2 1 Existing Dormitory Management System

Applications for rental management are growing rapidly providing resources that improve processes between landlords and tenants. By offering a user-friendly and reasonably priced platform specifically designed for the Thai market, DormiSpace fills in the remaining gaps. It has digital contracts, QR code payments, and OCR technology that reduces human error by automatically reading water and electricity meters. As well, the platform has a maintenance request feature that makes it convenient for tenants to use the app to report problems.

The main features of apps like Horganice, LivingBooker, and Yeeraf include room reservations, short-term property listings, and simple billing. Although they weren't created for long-term rental companies, these technologies help in improving the booking and advertising procedures. They lack crucial elements that dormitory landlords depend on for effective administration, like automated billing, monthly utility tracking, and maintenance reporting. On the other hand, more sophisticated features like online rent collecting, tenant data management, and usage analysis are provided by international platforms like TenantCloud, Abritel, RD Connect, and Ziprent. But their subscription costs are quite expensive.

Although there are several rental management systems available, just a few are designed for small dorm operators which demand a simple, reliable solution. DormiSpace enhances efficiency, reduces human error, and supports the digital transformation of the rental businesses by combining intelligent automation with essential management applications.

Comparison of Existing Applications

The comparison table illustrates an analysis of eight dormitory management platforms including Horganice, LivingBooker, Yeeraf, TenantCloud, Buildium, AppFolio, Resident Connect, and Zillow based on their main features and functions. It highlights the strengths and weaknesses of each platform to assess their suitability for different types of dormitory operations.

Table 3.1 Comparison Table of Existing Applications

Feature	Horganice	LivingBooker	Yeeraf	TenantCloud	Buildium	AppFolio	Resident Connect	Zillow	DormiSpace
User Management	✓	✓	✓	✓	✗	✓	✓	✓	✓
Room Management	✓	✓	✓	✓	✗	✓	✗	✓	✓
Utility Management	✓	✓	✓	✓	✗	✗	✗	✗	✓
Billing and Payment	✓	✓	✓	✓	✓	✓	✓	✓	✓
Analytics	✓	✓	✓	✓	✗	✓	✗	✗	✓
Notifications	✓	✓	✗	✓	✗	✗	✗	✗	✓
OCR image processing	✗	✗	✗	✗	✗	✗	✗	✗	✓
Digital Signature	✗	✗	✗	✓	✗	✗	✗	✗	✓

In comparison, we compare eight dormitory management platforms: Horganice, LivingBooker, Yeeraf, TenantCloud, Buildium, AppFolio, Resident Connect, and Zillow. The most comprehensive feature set, including room, billing, and contract management with local support, is provided by Horganice. Yeeraf and LivingBooker handle basic tasks, however they don't have complex tools like OCR. Due to cost and localization restrictions, TenantCloud, Buildium, and AppFolio are less appropriate for tiny Thai dormitories despite offering extensive features for large properties. Zillow primarily

facilitates property listings, whereas Resident Connect handles user management and billing.

2.3 Literature Review

The main technologies that support the DormiSpace rental management system are explored in this literature review. The technology reduces human error and replaces required manual recording by automatically reading water and electricity meters from uploaded images through Optical Character Recognition (OCR). In addition, it provides QR code payments for easy, convenient, and locally adapted transactions, and it includes digital and electronic signatures to make contract management simpler and more secure. Furthermore, real-time access, scalability, and data security are enhanced by cloud-based technologies like AWS and PostgreSQL. When combined, these technologies make DormiSpace an unique, effective, and user-friendly dorm and rental management system.

2.2.1 Rental Property Management Systems

Rental property management systems have evolved significantly over the last few decades, driven by the rapid adoption of digital technologies across industries [4]. The researchers highlighted that it was heavily reliant on manual, paper-based processes, which often led to inefficiencies, delays, and challenges in communication between landlords and tenants [5]. Landlords and property managers used physical documents to track tenant leases, maintenance requests, payments, and more. These manual systems were not only time-consuming but also prone to errors, making it difficult to provide a seamless tenant experience and efficiently manage multiple properties.

The authors of digital tools observed that the property management landscape has shifted toward automation and digitization [6]. Early innovations in property management software primarily focused on lease tracking, tenant communications, and rent collection. These solutions allowed landlords to automate basic administrative tasks, such as rent invoicing and tenant reminders, and facilitated digital storage of documents. However, these early systems often lacked comprehensive features, particularly for small landlords or family-owned

rental businesses, which often operated on smaller budgets and were not able to justify the expense of large enterprise systems [4], [6].

Recent advancements have seen the rise of cloud-based rental property management platforms that offer robust, scalable features. These systems now provide not only basic lease management but also integrated tools for payment processing, maintenance tracking, and document storage, streamlining operations for landlords and improving tenant satisfaction [7]. Tools such as Buildium, AppFolio, and Rentec Direct cater to medium and large-scale landlords by offering full-service solutions, including accounting, maintenance scheduling, and tenant communications, all integrated within a single platform. However, these solutions can be expensive and are often overkill for smaller-scale property managers who do not require such comprehensive features [7], [8].

In Southeast Asia, where mobile-first technology adoption is widespread, there is a growing demand for simple, affordable property management solutions tailored to the needs of smaller landlords. These systems prioritize ease of use, mobile accessibility, and regional payment integrations, such as QR payments [9]. In contrast to the more feature-heavy Western platforms, these systems often focus on simplifying communication and transaction management between landlords and tenants, such as rent collection and maintenance requests, with a particular emphasis on mobile solutions due to the high penetration of smartphones in the region.

Despite the growing variety of solutions, there are still significant gaps in the market for small landlords. Many existing platforms are either too complex or too expensive for small-scale property managers, and many fail to fully integrate payment options common in Southeast Asia, such as QR code payments, which are crucial in the region's digital economy. The rise of mobile-first, affordable platforms represents an opportunity for small landlords to gain access to automation, improving efficiency, reducing administrative workloads, and enhancing the tenant experience.

2.2.2 Digital Contracts and Signatures

The authors of Entrust explained that both Electronic Signatures (e-signatures) and Digital Signatures were widely used digital methods of authentication designed to replace traditional handwritten signatures [1]. The authors noted that e-signatures, which were often as simple as typing a name, clicking "Accept," or uploading a scanned version of a handwritten signature, offered key advantages in terms of convenience, speed, and efficiency. They stated that these features streamlined document workflows, reduced reliance on paper, and facilitated faster transactions. As a result, e-signatures were increasingly adopted across various industries, particularly those where quick processing and accessibility were essential, such as real estate, banking, and healthcare.

In contrast, Entrust reported that digital signatures employed Public Key Infrastructure (PKI), a cryptographic system that generated an electronic "fingerprint" linked to the signer's identity [1]. This method ensured that the document remained unchanged after signing and verified the signer's identity through the use of digital certificates. According to the authors, digital signatures offered greater security than e-signatures by providing stronger assurances of document integrity, authentication, and non-repudiation. These attributes made digital signatures especially suited for high-stakes environments where document authenticity and legal enforceability were critical, such as financial contracts or legal agreements.

Despite these security advantages, Entrust indicated that digital signatures tended to be more complex and costly to implement, particularly for small businesses with straightforward needs [1]. Their findings suggested that for organizations like DormiSpace, which aimed to implement an efficient and user-friendly property-management solution, e-signatures presented a more practical option. The authors explained that the ease of implementation and lower cost associated with e-signatures made them a suitable choice, providing a quick and effective way to facilitate contracts and agreements without requiring additional infrastructure or certifications.

2.2.3 Optical Character Recognition (OCR)

Optical Character Recognition (OCR) is a technology that enables the conversion of scanned or photographed images of text into machine-encoded text. Its development has been pivotal in digitizing paper-based documents and automating processes that were once manual. Early OCR systems primarily relied on pattern matching and template-based approaches. These systems worked by comparing the scanned image of each character against a database of known character shapes, essentially recognizing the text by matching it to predefined templates. While these systems were effective for standard fonts and printed text, they struggled with varying handwriting styles, distorted characters, and noisy image backgrounds. Despite these limitations, early OCR systems proved valuable for simple, high-volume tasks like digitizing printed books and documents in controlled environments.

As OCR technology evolved, more sophisticated feature extraction methods emerged, allowing for more flexible and accurate recognition of text. Modern OCR systems often employ machine learning techniques to recognize characters based on features such as edges, strokes, and patterns, rather than relying solely on template matching. This shift in approach allowed OCR systems to handle a wider range of fonts and handwriting styles. More recently, the integration of deep learning techniques, particularly convolutional neural networks (CNNs), has revolutionized OCR by enabling systems to learn and adapt to complex patterns in data, improving the accuracy of text recognition in both structured and unstructured documents. Deep learning models have significantly enhanced the recognition of text in images with variable fonts, lighting conditions, and noise, making modern OCR systems far more robust than their predecessors.

OCR technology has found widespread application across various industries, significantly contributing to the digitization of documents and automation of data entry. One of the most notable applications of OCR is in utility meter reading, where it is used to extract numerical data from scanned or photographed meter readings, eliminating the need for manual data entry. OCR is also commonly employed in ID verification systems, such as passport scanning or

automatic license plate recognition (ALPR), where it reads text from identification cards and documents for security and identification purposes. Additionally, OCR plays a crucial role in archiving historical documents and making them searchable by converting them into editable and indexable formats, which greatly enhances document accessibility and preservation.

Despite its advancements, OCR technology still faces several challenges that impact its effectiveness in certain scenarios. One of the primary challenges is image quality, as OCR systems are highly sensitive to distortions caused by poor lighting, low-resolution scans, or noisy backgrounds. Inaccurate image preprocessing can lead to errors in character recognition, particularly when dealing with faded text or skewed images. Handwriting recognition is another major challenge, as the variation in individual handwriting styles can cause inconsistencies in character interpretation. While some OCR systems have made strides in recognizing cursive and stylized handwriting, accuracy remains an issue, especially when faced with complex or non-standard handwriting. Additionally, multi-language support remains a challenge for OCR systems, as different languages have different character sets, syntax, and font styles. While OCR technology has made significant progress in supporting languages with Latin-based alphabets, recognizing non-Latin scripts (such as Arabic, Chinese, or Hindi) requires specialized models and datasets, making multilingual OCR systems more complex and less accurate.

2.2.4 Payment Processing in Digital Platforms

Industry reports explained that the evolution of online payments had drastically reshaped how businesses and consumers exchanged money. They stated that the payment ecosystem was initially dominated by credit and debit card transactions, and that the digital payment landscape later expanded to include digital wallets and QR-based payments, driven by the growth of mobile technology and changing consumer behavior [11], [12]. Early payment systems, according to these sources, primarily relied on card networks such as Visa and Mastercard, which facilitated global transactions but required physical cards and

were often perceived as slow or cumbersome for small, everyday transactions [11].

The literature noted that the rise of digital wallets such as Apple Pay, Google Pay, and Samsung Pay introduced a new level of convenience by allowing users to store card details securely on their mobile devices and make payments with a simple tap [12]. These studies argued that digital wallets paved the way for more seamless transactions, particularly in e-commerce and mobile payments. Researchers also highlighted that the latest shift in payment systems was the widespread adoption of QR-based payments, especially in regions like Southeast Asia, where QR codes had become a preferred method for making payments [11], [13]. QR codes were described as offering a simple, contactless solution for consumers and businesses by enabling payments without requiring specialized hardware such as card readers.

Various sources emphasized that security standards were essential for maintaining trust and preventing fraud in digital payment ecosystems. PCI DSS (Payment Card Industry Data Security Standard) was identified as the global benchmark for securing credit card transactions and ensuring that cardholder data was protected throughout the payment process [15]. Reports also stated that tokenization was widely used to safeguard sensitive card information by replacing it with a unique identifier or “token” that could not be used outside the specific transaction [16]. Meanwhile, 3D Secure (3DS) added an extra authentication layer by requiring cardholders to verify their identity before completing a transaction, reducing the risk of fraud [17]. With the rapid expansion of QR payments, cybersecurity researchers reported that secure QR protocols were developed to encrypt transaction data and ensure that payments were routed securely to the correct merchant [14].

In Southeast Asia, studies showed a significant surge in QR payment adoption, with countries such as Thailand, Malaysia, and India emerging as regional leaders [11], [13]. In Thailand, the Bank of Thailand reported that

PromptPay enabled users to make instant payments through QR codes linked to their bank accounts, eliminating the need for cash or physical cards [18]. Similar systems such as DuitNow in Malaysia and UPI in India were also found to support seamless real-time QR-based transactions. In China, platforms like Alipay and WeChat Pay were documented as dominating the payment landscape by integrating mobile wallets and QR payments into everyday activities, including shopping, transportation, and charitable donations [13].

The literature further indicated that global payment service providers such as Stripe and PayPal had established themselves as central players in the digital payment industry. Stripe was widely recognized for its developer-friendly API and ease of integration, enabling businesses of various sizes to accept online payments across multiple platforms [19]. PayPal, on the other hand, had built a strong reputation for secure and convenient cross-border transactions [20]. Within Southeast Asia, Omise (Opn) emerged as a prominent regional provider, offering robust support for QR payments and card transactions. Reports described Omise as particularly popular in Thailand, where it supported PromptPay and offered localized payment solutions for small and medium-sized enterprises [21].

Studies focusing on rental and property management systems highlighted that digital payments—especially QR-based systems—offered numerous benefits. They argued that QR payments provided lower transaction fees compared to traditional card networks, making them advantageous for small businesses like DormiSpace that may not be able to absorb higher processing fees. Researchers also indicated that QR-based payments enabled instant settlement, improving cash flow for landlords by reducing delays. Additionally, the familiarity of QR payments among tenants in regions such as Thailand further enhanced the user experience, as tenants were already accustomed to using mobile applications for everyday payments.

However, limitations were also identified in relation to QR payment systems. One major challenge commonly cited in the literature was reconciliation.

Unlike traditional credit card payments, where transactions could be automatically matched to specific payers, QR transactions sometimes required additional manual verification to link payments with the corresponding leases or tenants. Another challenge involved offline QR scanning—areas with poor internet connectivity could hinder QR code processing, leading to delays or missed payments. Researchers concluded that addressing these issues would be necessary for fully optimizing QR payment systems in rental platforms [14].

2.2.5 Notifications and Communication Technologies

Effective communication between tenants and landlords is crucial for maintaining smooth operations in rental property management. Timely notifications are essential for ensuring that both parties are informed about important updates, such as rent due dates, maintenance requests, or lease renewals. The role of notifications in tenant–landlord interactions cannot be overstated, as they enhance transparency, efficiency, and tenant satisfaction. For landlords, the ability to send instant reminders and updates not only saves time but also helps prevent issues such as late payments or missed maintenance requests. For tenants, receiving timely notifications about rent deadlines, maintenance scheduling, and other important matters improves their overall experience and fosters a sense of reliability and professionalism.

One of the most common methods for delivering notifications in modern applications is through push notification services. Push notifications allow applications to send messages to users even when they are not actively using the app. These notifications are a powerful tool for engaging users and ensuring they stay informed in real time. Several services facilitate push notifications, with Firebase Cloud Messaging (FCM) and Apple Push Notification Service (APNS) being two of the most widely used platforms. FCM, developed by Google, provides a cross-platform solution that supports notifications for both Android and iOS devices, making it particularly suitable for applications like DormiSpace, which targets users on both platforms. FCM is known for its ease of integration, scalability, and ability to send both push and in-app notifications. APNS, on the

other hand, is tailored for Apple devices, offering a more refined integration with iOS applications. While APNS is limited to iOS devices, it is the default choice for Apple-centric ecosystems and is renowned for its reliability and security features.

Numerous studies have explored the impact of timely communication and notifications on tenant satisfaction. Research suggests that proactive communication, such as sending reminders about upcoming payments, maintenance schedules, and policy changes, can significantly improve tenant satisfaction. In a study by Smith et al. (2020), it was found that tenants who received timely notifications about rent due dates and maintenance updates were 30% more likely to rate their rental experience as positive compared to those who did not receive such notifications. Similarly, a study by Johnson and Lee (2018) indicated that rental systems that utilized automated notifications had higher retention rates and lower complaint volumes, as tenants felt more informed and in control of their rental experience. These findings highlight the importance of effective communication tools in fostering positive tenant–landlord relationships.

Despite the clear benefits, there are challenges associated with push notifications. One of the main concerns is notification overload, where tenants may become frustrated if they receive too many notifications, especially if they are perceived as irrelevant or excessive. Personalization of notifications, where tenants receive only the most relevant updates, is crucial to mitigating this issue. Furthermore, push notifications require internet connectivity, and in areas with poor mobile network coverage, the effectiveness of notifications can be diminished. However, with proper design and user customization options, these challenges can be minimized, ensuring that notifications serve their intended purpose of improving communication and satisfaction without overwhelming the user.

2.2.6 Cloud-based Infrastructure in Small Business Applications

The shift from on-premise infrastructure to cloud computing has fundamentally transformed how businesses manage their IT resources. Traditionally, small businesses relied on physical servers and on-site data centers to store and process information, which often led to high upfront costs, ongoing maintenance expenses, and scalability limitations. Cloud computing, on the other hand, provides businesses with access to on-demand resources via the internet, allowing them to scale their infrastructure as needed without the capital investment and maintenance burdens associated with physical hardware. This shift has been particularly beneficial for small and medium-sized enterprises (SMEs), which typically have limited IT resources and require a cost-effective, flexible solution to meet their growing demands.

One of the key benefits of cloud computing is its ability to enable virtualization and containerization, two technologies that have revolutionized application deployment and management. Virtualization allows multiple virtual machines (VMs) to run on a single physical server, maximizing hardware utilization and reducing costs. Containerization, exemplified by tools like Docker, takes this concept further by isolating applications into lightweight containers that can be easily deployed and scaled across different environments. Docker provides a consistent and reproducible environment for developers, ensuring that an application behaves the same way in development, testing, and production environments. This level of consistency simplifies the deployment process and reduces the risk of environment-specific bugs, making it an ideal choice for small businesses that need to deploy and maintain applications efficiently[3].

In the context of property management, database technologies play a crucial role in handling large amounts of data related to tenant information, lease agreements, payments, and maintenance requests. The most commonly used databases in property management applications are relational databases like PostgreSQL and MySQL, which are particularly well-suited for structured data that requires ACID (Atomicity, Consistency, Isolation, Durability) properties. PostgreSQL is favored for its robustness, extensibility, and support for advanced

queries, while MySQL is known for its speed and simplicity[10]. However, as property management systems evolve to handle more complex, unstructured data (e.g., images, logs), NoSQL databases like MongoDB and Cassandra are becoming more common. These databases offer flexibility in handling varied data formats and scalability for handling large amounts of unstructured data, such as photos of rental properties, tenant communication logs, and maintenance records.

Cloud service providers like Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure have emerged as leading platforms for hosting applications and databases, offering a range of services tailored to the needs of SMEs. AWS is the most widely used cloud platform, providing comprehensive solutions for computing power (via EC2), storage (via S3), and database management (via RDS and DynamoDB). GCP offers competitive alternatives, particularly in the realm of machine learning and big data, while Azure is particularly popular among businesses that already use Microsoft products like Office 365 and Windows Server. All three platforms provide robust scalability, security, and compliance features that are essential for SMEs looking to expand their online presence and protect sensitive customer data. By leveraging cloud infrastructure, SMEs can scale their resources based on demand, reduce operational costs, and ensure data security without the need for large upfront investments in physical hardware.

For property management applications, these cloud platforms provide the reliability, security, and scalability that are critical for handling tenant data, processing payments, and ensuring business continuity. In addition to the infrastructure-as-a-service (IaaS) options, these platforms offer managed services for databases, networking, and application monitoring, making it easier for small businesses to manage their IT infrastructure with minimal expertise. The ability to scale on-demand and pay only for the resources used has made cloud hosting an attractive option for small property management businesses like DormiSpace, which may experience fluctuating usage based on the number of properties, tenants, and maintenance requests.

CHAPTER 3
METHODOLOGY

This chapter describes the methodology applied in developing the DormiSpace Application, a smart dormitory management system designed to support both land- lords and tenants. The objective of the system is to simplify property management processes such as room registration, rent payment, utility tracking, and maintenance requests through a single integrated platform. The methodology explains the overall workflow used during the project, including the system development approach, architectural de- sign, tools, and technologies employed to achieve the project objectives.

3.1 Overview of the Methodology

The development of DormiSpace followed a structured and iterative approach to ensure that all functional and non-functional requirements were met efficiently. The process began with requirement analysis and system design, followed by implementation, testing, and deployment. Throughout each phase, feedback from potential users and supervisors was collected to improve usability and reliability. To manage the development effectively, the Agile Software Development Method-ology—specifically the Scrum framework—was adopted. This approach emphasizes collaboration, flexibility, and continuous delivery of functional components within short timeframes known as sprints. By using Agile principles, the development team could prioritize essential features such as user authentication, room management, OCR-based utility tracking, and online payment integration while accommodating changes as the project progressed.

3.2 System Development Process

The project was divided into several stages, as illustrated in Figure ??. Each stage was designed to ensure that all components of DormiSpace were systematically analyzed, developed, and validated.

Figure 3.1 illustrates the iterative nature of the Agile process adopted in this project, showing how each phase—from requirement analysis to deployment—feeds back into the next sprint for continuous improvement.

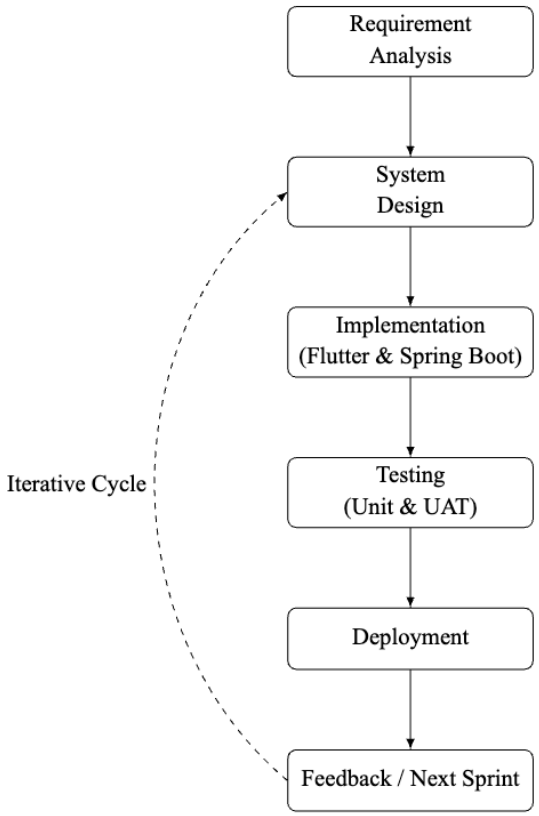

Figure 4.1: Agile-based system development process for the DormiSpace application.

3.2.1 Requirements Analysis

In this phase, both functional and non-functional requirements were gathered from the perspective of two user groups–landlords and tenants. Functional require-ments included account registration, property listing, rental contract management, pay-ment history tracking, and utility data input. Non-functional requirements focused on usability, security, and performance to ensure that the application remained stable and easy to use.

3.2.2 User Requirement Survey

Before designing the DormiSpace application, a user requirement survey was conducted to identify the essential features needed by both landlords and tenants. The

purpose of the survey was to understand the current challenges faced in dormitory management and to gather user expectations regarding system usability, functionality, and convenience.

A structured questionnaire was distributed to a sample group consisting of 27 participants, including 20 tenants and 7 landlords from different dormitories in Bangkok and nearby provinces. The survey contained both multiple-choice and open-ended questions covering the following topics:

• Current methods used for managing rent and utilities
• Frequency of communication issues between landlords and tenants
• Interest in digital payment options
• Preferences for mobile versus web-based access
• Desired additional features such as maintenance requests or utility bill tracking

The collected responses indicated that most tenants preferred a mobile-based system with integrated payment functions, while landlords emphasized the importance of automated record-keeping and monthly summaries. These findings were used to define the core functional requirements of DormiSpace, including the development of an OCR-based utility tracking feature and a secure payment gateway for rent and utility fees.

3.2.3 Survey Results and Analysis

The survey results revealed a strong demand from both tenants and landlords for a digital platform to simplify dormitory management. Tenants emphasized the importance of transparency and convenience, while landlords prioritized efficiency and financial oversight.

Among tenants, the most valued features were checking monthly utility bills (average rating 4.75), receiving bill notifications (4.70), reporting maintenance issues (4.70), and accessing rental contracts and documents (4.60). They also suggested additional functionalities such as real-time environmental updates, in-app chat for communication with landlords, and the ability to attach images when submitting maintenance requests.

Landlords rated features such as viewing historical financial data (5.00), tracking income and expenses (5.00), and using a dashboard with OCR-based meter reading (4.86) as most essential. They also requested integrations with smart meter APIs and automatic payment tracking.

Overall, the survey confirms a mutual demand for a centralized, transparent, and automated rental management system. These findings directly informed the feature design of DormiSpace, including the integration of OCR-based utility tracking, digital payments, maintenance reporting, and communication tools.

3.2.4 System Design

After the requirements were confirmed, the team designed the overall system structure using UML diagrams such as use case diagrams, class diagrams, and sequence diagrams. These diagrams defined the relationship between system components and guided the implementation of backend services and user interfaces.

3.2.5 Implementation

During this stage, both frontend and backend components were developed in parallel. The frontend was implemented using the Flutter framework with Dart programming language, while the backend was built using Spring Boot with RESTful API architecture. The backend was connected to a PostgreSQL database, and both modules communicated through secure HTTP requests. An OCR (Optical Character Recognition) module was also developed to read utility meter images and automatically record usage data. The application integrated a payment gateway using Omise API to process rent and utility payments securely.

3.2.6 Testing and Validation

Multiple testing strategies were applied, including unit testing with JUnit, API testing using Postman, and user acceptance testing (UAT) with sample users. These tests

verified that the application met its functional requirements and ensured proper data handling, user role management, and accuracy of OCR readings.

3.2.7 Deployment

Finally, the application was deployed in a production-like environment. The backend services were hosted on a local server during development and later deployed to a cloud-based environment for public access. The mobile application was built into an Android package (APK) and tested across multiple devices for performance validation.

3.3 System Architecture Design

The architecture of DormiSpace follows a three-tier client-server model consisting of the presentation layer, application layer, and database layer. This structure enables modularity, scalability, and easier maintenance, as shown in Figure 4.2.

Figure 4.2: Overall system architecture of the DormiSpace application.

3.3.1 Presentation Layer

The presentation layer was implemented using Flutter, providing a responsive user interface for both landlords and tenants. This layer handles user interactions such as property registration, utility input through camera capture, and payment confirmation. It communicates with the backend services through RESTful API calls.

3.3.2 Application Layer

The application layer was built using Spring Boot, which handles business logic, request routing, and API responses. It manages key modules such as user authentication, property management, utility processing, and payment transactions. Each module was implemented as a separate controller-service-repository structure to promote modularity and maintainability.

3.3.3 Database Layer

The PostgreSQL database stores all persistent information, including user profiles, property details, utility records, and payment history. Entity-relationship (ER)

diagrams were designed to represent the relationships between these data entities, ensuring referential integrity and optimized query performance.

3.3.4 External Services

External services such as the Omise payment gateway and OCR engine were integrated to extend system functionality. The OCR service was used to extract numeric values from utility meter images, while the Omise API enabled secure credit card and PromptPay transactions. All external API communications were encrypted using HTTPS protocols to protect user data.

3.4 Summary

This chapter presented the methodology employed in developing the DormiSpace application, detailing the development model, system design, and implementation processes. The next chapter discusses the system implementation and testing results in detail.

CHAPTER 5
EVALUATION AND RESULTS

CHAPTER 6
CONCLUSION AND FUTURE WORK

REFERENCES

[1] Entrust. Digital Signature vs. Electronic Signature. Entrust Resource Center [Online]. 2024 [cited 2025 Sep. 21]. Available from: https://www.entrust.com/resources/learn/digital-signature-vs-electronic-signature. [Accessed: Nov. 10, 2025].

[2] IBM. Optical Character Recognition (OCR). IBM Think [Online]. 2024 [cited 2025 Sep. 21]. Available from: https://www.ibm.com/think/topics/optical-character-recognition. [Accessed: Nov. 10, 2025].

[3] Nadia Reckmann. What Is Cloud Computing? Business News Daily [Online]. 2024 [cited 2025 Sep. 21]. Available from: https://www.businessnewsdaily.com/4427-cloud-computing-small-business.html. [Accessed: Nov. 10, 2025].

[4] MarketGrowthReports, Property Management Software Market Size & Share Trends, 2033. [Online]. Available: https://www.marketgrowthreports.com/market-reports/property-management-software-market-105282. [Accessed: Nov. 19, 2025].

[5] 302 Properties, "The Evolution of Property Management Technology Over the Last Decade," 302Properties, [Online]. Available: https://302properties.com/the-evolution-of-property-management-technology-over-the-last-decade/. [Accessed: Nov. 19, 2025].

[6] TurboTenant, "Rent Management Software," TurboTenant, [Online]. Available: https://www.turbotenant.com/rent-collection/rent-management-software/. [Accessed: Nov. 19, 2025].

[7] Primior, “Why Smart Property Managers Are Switching to Modern Property Management Tools in 2025,” Primior, [Online]. Available: https://primior.com/why-smart-property-managers-are-switching-to-modern-property-management-tools-in-2025/. [Accessed: Nov. 19, 2025].

[8] Buildium, “Buildium,” Buildium, [Online]. Available: https://www.buildium.com/. [Accessed: Nov. 19, 2025].

[9] Tech Collective SEA, “PropTech in Southeast Asia,” TechCollectiveSEA, Jul. 10, 2023. [Online]. Available: https://techcollectivesea.com/2023/07/10/proptech-in-southeast-asia/. [Accessed: Nov. 19, 2025].

[10] Ziga Kokalj. PostgreSQL vs. MySQL: What’s the Difference? DataCamp Blog [Online]. 2024 [cited 2025 Nov. 2]. Available from: https://www.datacamp.com/blog/postgresql-vs-mysql. [Accessed: Nov. 19, 2025].

[11] Statista, “Digital payments in Southeast Asia – statistics & facts,” Statista, 2024. [Online]. Available: https://www.statista.com/topics/9451/digital-payments-in-southeast-asia/. [Accessed: Nov. 19, 2025].

[12] HKIFOA, “Digital wallets and payment trends in Asia-Pacific,” HKIFOA Report, 2024. [Online]. Available: https://hkifoa.com/wp-content/uploads/2024/12/digital-wallets-and-more-asia-pacific.pdf hkifoa.com [Accessed: Nov. 19, 2025].

[13] J. Allan, “QR Codes and Smartphones Propel Asia’s Cashless Revolution,” Thailand Business News, Feb. 13, 2025. [Online]. Available: https://www.thailand-business-news.com/banking/193902-qr-codes-and-smartphones-propel-asias-cashless-revolution?.com. [Accessed: Nov. 19, 2025].

[14] HITRUST, “Securing Southeast Asia’s QR Payments: Key Security Challenges and Solutions,” HITRUST, Jan. 21, 2025. [Online]. Available:

https://www.hitrust.com/blog_Securing%20Southeast%20Asia%27s%20QR%20Payments_Key%20Security%20Challenges%20and%20Solutions.html?com. [Accessed: Nov. 19, 2025].

[15] PCI Security Standards Council, "PCI DSS Standards Overview," PCI SSC. [Online]. Available: https://listings.pcisecuritystandards.org/pdfs/pcissc_overview.pdf?.com. [Accessed: Nov. 19, 2025].

[16] Mastercard Newsroom, "What is Tokenization?", Mastercard Newsroom, Mar. 19, 2024. [Online]. Available: https://newsroom.mastercard.com/news/perspectives/2024/what-is-tokenization/?.com. [Accessed: Nov. 19, 2025]

[17] PCIpal, "Protecting Online Card Payments: Explore PSD2 and 3D Secure," PCI Pal, 2024. [Online]. Available: https://www.pcipal.com/knowledge-center/resource/protecting-online-card-payments-explore-psd2-and-3d-secure/?.com. [Accessed: Nov. 19, 2025]

[18] Bank of Thailand, "Payment Systems Annual Report, 2021," Bank of Thailand, 2021. [Online]. Available: https://www.bot.or.th/content/dam/bot/documents/en/research-and-publications/reports/payment-report/payment-systems-annual-report/Payment_2021_E.pdf?.com. [Accessed: Nov. 19, 2025]

[19] Stripe, "Global Payments Platform," Stripe, 2024. [Online]. Available: https://stripe.com/en-th/payments?.com. [Accessed: Nov. 19, 2025]

[20] Leceipt, "รู้จักเพย์พาล (PayPal) ว่าคืออะไร มีความสำคัญ และการทำงานอย่างไร?", Leceipt Blog, Dec. 09, 2022. [Online]. Available: https://www.leceipt.com/blog/%E0%B9%80%E0%B8%9E%E0%B8%A2%E0%B9%8C%E0%B8%9E%E0%B8%B2%E0%B8%A5-paypal-%E0%B8%84%E0%B8%B7%E0%B8%AD%E0%B8%AD%E0%B8%B0%E0%B9%84%E0%B8%A3 leceipt.com [Accessed: Nov. 19, 2025]

[21] Opn (formerly Omise), "Omise Payments and QR Support," Opn Payments, 2024. [Online]. Available: https://www.omise.co/payment-methods/thailand. [Accessed: Nov. 19, 2025]

AI USAGE ACKNOWLEDGMENT STATEMENT

We acknowledge that we have utilized Generative Artificial Intelligence (Generative AI) tools (….., ……, …...) in preparing this work. The AI tools were used for the following purposes:

- o Idea development
- o Research assistance (e.g., literature review)
- o Summarization and paraphrasing
- o Draft generation
- o Language correction, grammar checking, or translation
- o Code editing and debugging
- o Code generation
- o Employing AI in the final product (e.g., generative AI API, etc.)
- o Other (please specify): ……..

APPENDIX A

<INSERT YOUR TOPIC>